Python Programming Foundations

Week 1 - First Programs and Basic Values

Day 3: "Basic Values Together"

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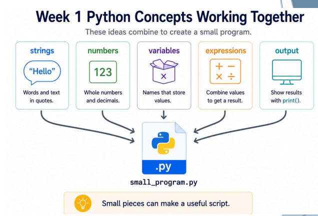
Put The Pieces Together

This week, you learned the smallest useful Python pieces.

Today, you combine them into tiny programs that run and make sense.



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Quick Review

Monday made output visible.

Tuesday made numeric value flow visible.

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What Counts as Success Today

NOT a large program

Small Program that:

- Runs
- Produces output
- Can be Explained

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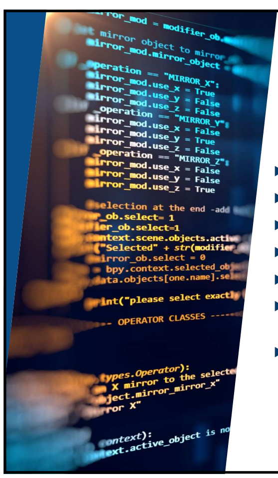

What We Will Use Today

- strings
- numbers
- variables
- simple expressions
- `print()`
- optional `input()` if stable
- basic type awareness

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Working Set

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What to “Skip” for Now

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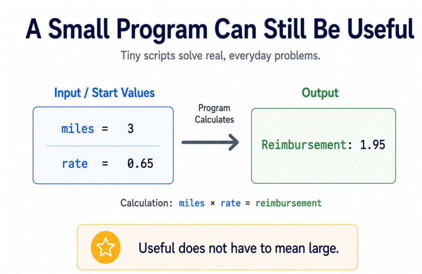

A useful first program does not have to be large.

It only needs a clear purpose, visible output, and values you can explain.

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Combine Values Carefully

Your program may use text and numbers together.

The important question is whether you can explain each value's role.

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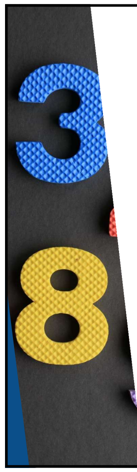

Type Awareness Through Use

For Week 1, recognize:

- Text
- Whole numbers
- Decimal numbers
- True/false values (boolean)

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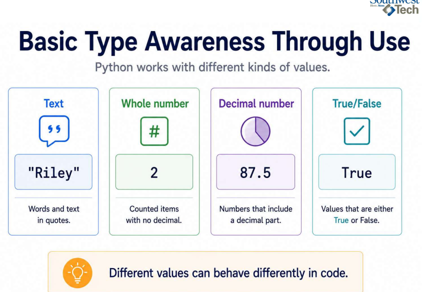

Type Awareness

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Explain Every Main Line

Before submitting Assignment 1, ask: can I explain every main line?

If a line works but you cannot explain it, slow down and inspect it.

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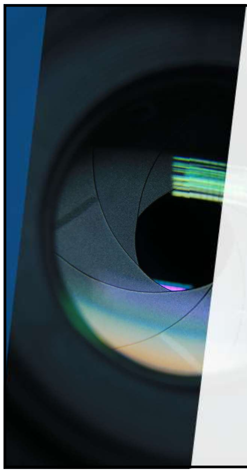

Demo #1

"Basic Type Awareness"

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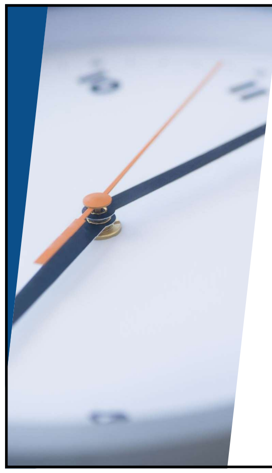

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What to watch for

As the demo runs, watch for two things:

- How the program stores several different kinds of values.
- Recognition. This is the goal - ***not*** a full type inventory.

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Basic Type Awareness Demo

Every variable holds a value, and each value has a kind.

Variable Name	Value	Kind of Value
student_name	"Riley"	text
assignments_completed	2	whole number
average_score	87.5	decimal number
is_passing	True	true/false

Recognize the value before you explain the program.

Type Demo

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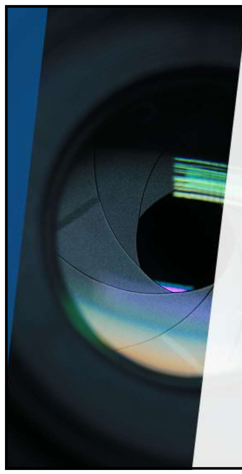

Optional Demo #2

"Input And Number Conversion"

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Optional Demo: Input And Number Conversion

- When users type into `input()`, Python receives text first.
- For math, the text must be converted into a number.

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Typing vs. Copy-Paste

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Learning Move:
Type It When It Is New

Typing code can be part of learning.

Copying is not automatically wrong, but it should not replace understanding.

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Typing it Out

Typing code manually slows the process down in a useful way.

When you type an example, you are not just moving characters onto the screen. You are giving your mind time to process:

- the order of the instructions
- the names of variables
- the structure of the logic
- the punctuation and syntax
- the relationship between one line and the next
- what the program is actually doing

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Copying and Pasting

Copying and pasting can also be appropriate. It may make sense when:

- the code is long
- the concept has already been learned
- the focus is on a different part of the task
- you understands the logic already
- the code is being used as a scaffold for something else

Copy/paste becomes a problem when it replaces understanding. - Before copying, You should ask:

- Do I understand what this code does?
- Do I need to understand this more deeply right now?
- Am I copying because it saves time, or because I am avoiding thinking?

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Connection to AI Prompting

Typing a prompt is not just the act of entering text into a tool. It is often part of the thinking process.

As a person writes a prompt, they may:

- clarify the goal
- revise wording
- discover missing constraints
- notice ambiguity
- reshape the request
- improve the structure of their own thinking

This is closely connected to AI-assisted development and Refraction-Based Architecture.

- The human is not merely asking AI for output - ***The human is shaping intent!***

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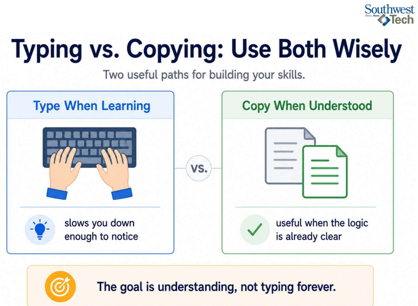

Typing vs Copy-Pasting

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Common Failure:
Too Much Is Too Much

Trying to impress can make the first assignment harder than it needs to be.

Small, clean, and explainable is better than large and confusing.

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Common Failure: Working But Unexplained

- Code that runs is not the whole goal.
- You also need to explain what values the program uses and what output it creates.

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From Demo to Lab

For Assignment 1:

- Finish two or three tiny programs.
- Each program should run, show output, and make sense when you explain it.

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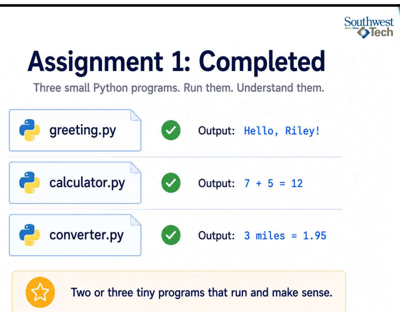

Assignment Completion

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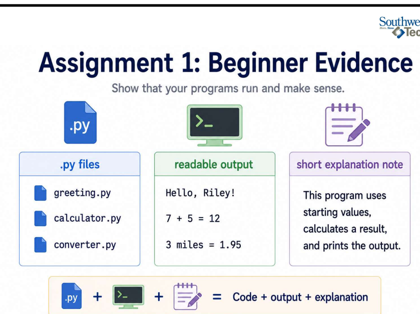

Assignment Evidence

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Lab On Success Check

- Week 1 worked if your small programs run and you can explain them.
- Understanding what the code is doing:
  - What values does (each) program use?
  - What output does it produce?
  - Where does a value change?
  - Could you explain this script line by line to someone else?

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Thank you!

Have Fun!

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